

A Counterexample to Arrhenius's Sixth Impossibility Theorem

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Abstract

Gustaf Arrhenius's sixth impossibility theorem in population ethics, the one he regards as resting on the weakest and most compelling conditions, says that no population axiology satisfies Egalitarian Dominance, General Non-Extreme Priority, Non-Elitism, Weak Non-Sadism, and the Weak Quality Addition Condition. Teruji Thomas has observed that the published proof contains a gap, and Elliott Thornley has restated the last condition with two quantifiers reversed so that the proof goes through. I show that the gap cannot be closed: the theorem as Arrhenius states it is false. There is a complete, transitive welfarist axiology, defined on Arrhenius's own integer-indexed welfare levels, that satisfies all five of his conditions in their exact formulations. The axiology ranks populations first by whether their misery exceeds what their excellent lives can offset, and otherwise by total welfare. It accepts the Repugnant Conclusion and rejects the Very Repugnant Conclusion, which is precisely the combination the theorem was designed to rule out. The repaired theorem is true, and I show what its extra premise buys: the four remaining conditions force only a background-relative Very Repugnant Conclusion, and the strengthened condition is exactly what excludes axiologies on which the quantity of misery that lives barely worth living can never make up for depends on how much excellence the world already contains. Every result stated as a theorem here, including the repaired theorem and Thomas's reconstructions of the other five, is verified in the Lean 4 proof assistant; the counterexample was found by a method, described here, that systematically searches for such axiologies.

I The theorem

Population ethics has a body of impossibility theorems that do for the comparison of differently sized populations what Arrow's theorem did for voting. Each lists a handful of conditions on a betterness ordering of populations, each condition apparently undeniable on its own, and shows that no ordering satisfies them all. The problem they formalize is Parfit's (1984, ch. 17; 2016): that the ways of avoiding the Repugnant Conclusion each seem to cost something no less repugnant. The first such theorems were proved by Ng (1989) and Blackorby and Donaldson (1991); Gustaf Arrhenius has since proved six (Arrhenius 2000b, 2000a, 2003, 2009, 2011), which are now the standard reference point (Greaves 2017), and the sixth is the one he regards as best, because its conditions are logically the weakest. It says:

The Sixth Impossibility Theorem. There is no population axiology which satisfies the

Egalitarian Dominance, the General Non-Extreme Priority, the Non-Elitism, the Weak Non-Sadism, and the Weak Quality Addition Condition. (Arrhenius 2011, 9)

The theorem has been treated as settled. It is stated as a result in Thornley's work on population prospects (Thornley 2021), discussed as such by Carlson (2022), and it is the version of Arrhenius's results that contributors to the Oxford Handbook of Population Ethics take as their point of departure (Arrhenius et al. 2022); Arrhenius's book, which collects the six theorems, has long circulated in draft (Arrhenius, forthcoming), and I do not know whether its current version states the condition as the 2011 chapter does. Two readers have noticed that something is wrong with the proof. Teruji Thomas, in an unpublished reconstruction of all six theorems, found that the published argument establishes only a weaker conclusion than the one Arrhenius states, and he noted in print that "the proof of his favoured sixth theorem contains an error in the derivation of the 'Restricted Quality Addition Condition' (Arrhenius, 2011, Lemma 1.3). As far as I know, one has to slightly strengthen the premisses of the theorem, in a way unlikely to cause further controversy" (Thomas 2018, n. 4; cf. Thomas 2016). Elliott Thornley, following Thomas, restates the Weak Quality Addition Condition with its first two quantifiers reversed, remarks that "the Sixth Impossibility Theorem actually requires the slightly stronger condition stated here," and adds that "the stronger version remains a compelling adequacy condition" (Thornley 2021, n. 7).

Thomas's remark already implies that the theorem, and not only its proof, has to change; but neither he nor Thornley shows that it does, and the field has gone on citing the original statement with the strengthening treated as a formality. I will show that the strengthening is not a formality. The theorem as Arrhenius states it is false. There is a complete, transitive, anonymous welfarist axiology, defined on exactly the integer-indexed welfare levels that Arrhenius's framework assumes, that satisfies all five of his conditions in the exact formulations he gives them. I call it the *capacity view*. It ranks populations first by whether the misery they contain exceeds what their excellent lives can offset, and, when it does not, by total welfare. It accepts the Repugnant Conclusion, as total utilitarianism does, and it rejects the Very Repugnant Conclusion, which is the combination of attitudes that the move from Arrhenius's earlier theorems to the sixth was designed to make untenable. So the sixth theorem does not refute the position it was built to refute.

The strengthened condition that Thomas and Thornley supply does make the theorem true; Thomas proved the strengthened theorem in his note, and I have checked his proof. What I want to add is an account of what the strengthening buys, since the capacity view makes the difference between the two conditions concrete. In its original form the Weak Quality Addition Condition says that, for any population, there is some quantity of very bad lives such that adding those lives together with any number of lives barely worth living is no better than adding some number of excellent lives. The strengthened condition fixes the quantity of very bad lives first and then quantifies over populations. The difference is whether the amount of misery that lives barely worth living can never make up for is allowed to depend on how much excellence the population already contains. The capacity view says that it is. The four other conditions, I will show, cannot force it not to be: what they entail is a Very Repugnant Conclusion relative to a background that may be chosen after the quantity of misery is

fixed, and nothing stronger. The repaired theorem is therefore a theorem about background-relative repugnance, and whether its new premise is as “compelling” as the old one is a question about whether capacity views deserve to be ruled out by an adequacy condition rather than by argument.

The paper also has a methodological point. The counterexample was not found by inspection. It was found by looking for a quantity that none of the four non-conclusion conditions can decrease, and then building an axiology that gives that quantity lexical priority. §5 describes the method, which applies to any impossibility theorem whose conditions are “transfer” conditions of the kind Arrhenius uses, and which turns the search for countermodels into a routine check. Everything in the paper has been verified in the Lean 4 proof assistant: that the capacity view satisfies the five conditions, that it violates the strengthened condition, and that Thomas’s reconstructions of Theorems 1 through 5 and of the corrected sixth theorem are valid. §6 describes the verification, and the source is available with the paper.

2 The framework and the five conditions

Arrhenius’s framework is this (Arrhenius 2011, 4–7). A life has a welfare level; welfare levels are quasi-ordered; and there is a neutral level, with lives above it worth living and lives below it worth not living. Arrhenius assumes *Discreteness*: between any two welfare levels there are only finitely many others. (He argues that nothing turns on this, since if welfare is dense one can pass to a discrete subset of levels, as fine-grained as one likes, on which the conditions remain plausible (Arrhenius 2011, 6). I will return to the point in §4.) Given Discreteness, welfare levels can be indexed by integers, with 0 the neutral level, so that the level $w + 1$ is “slightly higher” than w . A *population* is a finite set of lives. Following Thomas (2016) I will identify a population with its welfare distribution, the function that assigns to each level the number of lives at that level, and write $n \cdot x$ for a population of n lives all at level x , and $P + Q$ for the union of two populations.¹ An *axiology* is a relation “at least as good as” on populations, assumed to be reflexive and transitive but not necessarily complete; “better than” is the strict part.

Three further parameters appear in the conditions. There is a “very high” welfare level, which I write A ; a “very low positive” range of levels, which Arrhenius takes to be $R(1, y)$, the levels from 1 up to some y ; and a “very negative” level. In the exact formulations these are not fixed in advance but are quantified over inside each condition, which is one of the things that makes the formulations exact; a condition that says “there is a very high level u such that...” does not presuppose that we know where “very high” begins. The reader who wants the informal versions will find them alongside the exact ones in Arrhenius (2011, 7–9); I give the exact ones, in distribution notation, since the result depends on them. All welfare levels are integers, all population sizes are non-negative integers, and ranges $R(x, y)$ consist of at least three levels.

¹Arrhenius’s populations are sets of particular lives, and his conditions are stated with the rider “other things being equal,” which abstracts from non-welfare differences between them. Since every condition is invariant under replacing lives by others at the same level, nothing is lost by working with distributions. Arrhenius’s own exact formulations use the notation $A \subset W_x$, $N(A) = n$ for what I write $n \cdot x$.

Egalitarian Dominance (ED). If $N(A) = N(B)$, every member of B has welfare below x , and $A = n \cdot x$, then A is better than B .

General Non-Extreme Priority (GNEP). For any level z there are a positive level u , a positive range $R(1, y)$ with $u > y$, and a number $n > 0$ such that for any $x \geq u$, any $B \subset R(1, y)$ with $N(B) = n$, and any population E : $n \cdot x + 1 \cdot z + E$ is at least as good as $B + 1 \cdot (z + 1) + E$.

Non-Elitism (NE). For any levels x, y with $x - 1 > y$ there is $n > 0$ such that for any $D \subset R(y, x)$: $(n + 1) \cdot (x - 1) + D$ is at least as good as $1 \cdot x + n \cdot y + D$.

Weak Non-Sadism (WNS). There are a level $x < 0$ and a number n such that for any level $y > 0$, any $B \subset W_y$, and any population C : $B + C$ is at least as good as $n \cdot x + C$.

Weak Quality Addition (WQA). For any population X there are a level $x < 0$, positive ranges $R(u, v)$ and $R(1, y)$ with $u > y$, and numbers $n, m > 0$ such that for any $z \geq u$ and any $B \subset R(1, y)$: $n \cdot z + X$ is at least as good as $B + m \cdot x + X$.

The informal content is familiar. ED says that a perfectly equal population is better than any same-sized population every member of which is worse off than its members. GNEP says that lowering one person's welfare by a step is never categorically worse than failing to raise a fixed number of others from very low positive to very high welfare, whatever else is true of the population. NE says that one person's small loss can always be compensated by a sufficient number of others' gains from a lower level to the same level she falls to, at least when the rest of the population lies between those levels. WNS says that there is some number of lives at some very negative level whose addition is no better than the addition of any number of lives with positive welfare. And WQA says that for any population there is some quantity of very bad lives such that adding them, together with any number of lives barely worth living, is no better than adding some number of very good lives.

Three features of WQA matter below. First, the conclusion compares additions to a *background* X , and the witnesses (x, m, u, y, n) are chosen after X is given: the condition begins "for any population X there are." Second, B ranges over all populations in the low range, of every size, so that the condition says the m bad lives cannot be compensated by any number of marginal lives. Third, the condition is a weakening of Arrhenius's Quality Addition Condition, which drops the m bad lives and says that for any X some number of very good lives added to X is at least as good as any number of marginal lives added to X . Quality Addition is the denial of what Thomas calls Repugnant Addition, a background-relative form of the Repugnant Conclusion, and total utilitarians reject it. Weak Quality Addition is the denial of a background-relative form of the Very Repugnant Conclusion, that for any population of excellent lives there is a better population consisting of some very bad lives together with enough lives barely worth living. Arrhenius weakened the condition because the Very Repugnant Conclusion is, as he and his coauthors put it, "intuitively more compelling" as a thing to avoid than the Repugnant Conclusion (Arrhenius et al. 2021, 119): it is meant to bind even those who, like Tännsjö (2002) and Huemer (2008), have made their peace with the Repugnant Conclusion, a group that has lately grown (Zuber et al. 2021) as it has become clear how little is needed to derive the

conclusion (Spears and Budolfson 2021).

3 The capacity view

Fix a “very high” level $A \geq 4$, so that the very low positive levels 1, 2, 3 lie strictly below it. For a population P let

- $H(P)$ be the number of lives in P at or above level A (the *excellent* lives);
- $T^-(P)$ be the total welfare of the lives in P with negative welfare, a non-positive number;
- $T(P)$ be the total welfare of P .

Let $I(P) = H(P) + T^-(P)$, and for a fixed *baseline capacity* $c \geq 0$ let $J(P) = \min(0, I(P) + c)$. The quantity I measures the population’s misery against its excellence, counting each excellent life as able to offset one unit of negative welfare; c is an amount of misery that any population, including the empty one, can absorb. J is zero when the excellent lives, together with the baseline, cover the misery, and negative, equal to the uncovered misery, when they do not. The *capacity view* (with parameters A and c) ranks populations lexically by (J, T) :

P is at least as good as Q if and only if either $J(P) > J(Q)$, or $J(P) = J(Q)$ and $T(P) \geq T(Q)$.

This is a complete and transitive ordering, since it is the lexicographic order on pairs of integers. It is welfarist and anonymous, since it depends only on the distribution. In words: a population whose misery exceeds its capacity to redeem it is worse than any population whose misery does not, and among populations whose misery exceeds their capacity the one with the smaller excess is better; among populations within capacity, total welfare decides. The exchange rate of one excellent life per unit of misery is a choice of convenience; any positive weight on H works, with the witness n in GNEP and NE adjusted. The simplest member of the family has $c = 0$, and the reader may take that as the official countermodel; the baseline matters only for the discussion in §4.

The view has some features its critics will want to know about at once. It is lexical, and so discontinuous: a single unit of uncovered misery outweighs any quantity of total welfare. With $c = 0$ this has a consequence worth stating in its starkest form: a population of a billion lives at level 3 together with one life at level -1 has $J = -1$, and is therefore worse than the empty population. Lives below the very high level cannot redeem misery at all, however many there are. With $c \geq 1$ the consequence disappears, since the baseline absorbs the one unit of misery, and I return to what this shows in §4. It implies the Repugnant Conclusion, since populations containing no misery are ranked by total welfare, so that for any n lives at level A there is a larger population of lives at level 3 that is better. It rejects the Very Repugnant Conclusion: for $m|z| > c$, the population $m \cdot z + N \cdot 3$ has $I + c = mz + c < 0$ and so is worse than $n \cdot A$, which has $J = 0$, for every N . And it is not meant as a proposal about how to rank populations. It is a proof that the five conditions can be jointly satisfied, and its interest lies in what it reveals about them.

Theorem 1. For every $A \geq 4$ and $c \geq 0$, the capacity view satisfies ED, GNEP, NE, WNS and WQA

in the exact formulations of §2.

I give the arguments at the level of detail a reader needs to see why each condition holds; the fully formal versions are the Lean proofs described in §6. The key observation is that every one of the first four conditions compares two populations that differ by a *transfer*: some lives are moved between levels, or some are added. Write ΔI and ΔT for the change in I and T from the population the condition says is no better to the one it says is at least as good. If $\Delta I \geq 0$ and $\Delta T \geq 0$ the capacity view agrees with the condition, since J is monotone in I (the baseline c cancels from every such comparison); and if $\Delta I \geq 0$ and $\Delta T > 0$ it agrees with the strict version. Background populations drop out, because I and T are additive over union.

Egalitarian Dominance. $A = n \cdot x$ and every member of B is below x , with $N(B) = n$. Then $T(A) = nx > T(B)$. For I : if $x \geq A$ then $H(A) = n \geq H(B)$, and if $x < A$ then $H(B) = 0 = H(A)$; and $T^-(A) \geq T^-(B)$, since if $x < 0$ every member of B is negative and below x , so $T^-(B) = T(B) < nx = T^-(A)$, while if $x \geq 0$ then $T^-(A) = 0 \geq T^-(B)$. So $\Delta I \geq 0$ and $\Delta T > 0$.

General Non-Extreme Priority. Given z , choose $u = A$, $y = 3$, $n = 2$. The condition compares $B + 1 \cdot (z + 1) + E$ with $2 \cdot x + 1 \cdot z + E$, where $x \geq A$ and B is two lives in $\{1, 2, 3\}$. The two excellent lives raise H by 2; the person who moves from $z + 1$ to z lowers H by 1 if $z + 1 = A$, and lowers T^- by 1 if $z + 1 \leq 0$; B contributes nothing to H or T^- . So $\Delta I \geq 2 - 1 - 1 = 0$. And $\Delta T = 2x + z - T(B) - (z + 1) \geq 2A - 6 - 1 > 0$.

Non-Elitism. Given $x - 1 > y$, choose $n = 1$. The condition compares $1 \cdot x + 1 \cdot y + D$ with $2 \cdot (x - 1) + D$, for $D \subset R(y, x)$. $\Delta T = x - 2 - y \geq 0$. For I there are cases. If $x - 1 \geq A$ the two lives at $x - 1$ are excellent and $\Delta H \geq 0$, while $\Delta T^- \geq 0$ since only the life at y can be negative and it is raised. If $x < A$ then $\Delta H = 0$ and a short calculation shows $\Delta T^- \geq 0$: the only case needing care is $x = 0$, where two lives at -1 replace one at 0 and one at $y \leq -2$, and $\Delta T^- = -2 + |y| \geq 0$. The remaining case is $x = A$, where one excellent life is lost: $\Delta H = -1$. If $y < 0$ then the life raised from y gives $\Delta T^- = |y| \geq 1$, so $\Delta I \geq 0$. If $y \geq 0$ then $D \subset R(y, A)$ contains no negative lives, so both populations have $T^- = 0$, both have $I + c \geq 0$, both have $J = 0$, and the comparison is decided by T , which does not decrease. This is the one place where I can fall under a licensed transfer, and it can fall only in a population whose J is already zero and stays zero. The restriction on D in Non-Elitism is what makes this work, and it is the restriction that Arrhenius's General Non-Elitism Condition, used in his fifth theorem, drops; the capacity view violates General Non-Elitism, as it must, since the fifth theorem is valid.

Weak Non-Sadism. Choose $x = -1$ and $n = 1$. The condition compares $1 \cdot (-1) + C$ with $B + C$, where B is any number of lives at some $y > 0$. $\Delta I = H(B) + 1 \geq 1$ and $\Delta T = T(B) + 1 \geq 1$.

Weak Quality Addition. Given X , choose $x = -1$, $u = A$, $y = 3$, $n = 1$, and m large enough that $m > I(X) + c$; concretely $m = \max(0, I(X) + c) + 1$. For any $z \geq A$ and any $B \subset R(1, 3)$: $I(1 \cdot z + X) = 1 + I(X)$, while $I(B + m \cdot (-1) + X) + c = I(X) + c - m < 0$, since B contributes nothing to I . So $J(B + m \cdot (-1) + X) = I(X) + c - m < \min(0, 1 + I(X) + c) = J(1 \cdot z + X)$,

and $1 \cdot z + X$ is better than $B + m \cdot (-1) + X$ for every B , whatever its size.

The last argument displays the mechanism. Weak Quality Addition lets the quantity of misery depend on the background, and the capacity view uses exactly that freedom: it chooses enough misery to exceed the background's capacity, after which no quantity of marginal lives can help, because marginal lives contribute to T and not to I . The four other conditions never require the view to compare a population that has exceeded its capacity favourably with one that has not, because, as the case analysis shows, none of the transfers they license lowers I except in populations with no misery at all.

Since the capacity view satisfies the five conditions, the Sixth Impossibility Theorem is false. Arrhenius's argument, I should stress, is not invalid at some obscure step; it is invalid at exactly the step Thomas identified. In the proof of his Lemma 1.3, which derives a Restricted Quality Addition Condition from Weak Quality Addition and an intermediate condition, step (12) applies Weak Quality Addition to the background population $A_1 \cup X$ using witnesses that the condition supplies for the background X (Arrhenius 2011, 18–19). The population A_1 consists of excellent lives whose number depends, through the intermediate condition, on those very witnesses. On the capacity view the step fails for a simple reason: adding A_1 raises the background's capacity, and the quantity m of misery that sufficed to exceed the capacity of X no longer exceeds the capacity of $A_1 \cup X$, so that in the enlarged background the marginal lives win.

4 The repair and what it costs

Thomas states the conclusion that the four conditions do yield, and Thornley states the condition whose denial it is. In Thomas's notation (2016, 11), the conclusion Arrhenius intended is

Very Repugnant Addition (VRA). For some population P , any level $z < 0$, and any $m, n \geq 1$, there is N such that $P + m \cdot z + N \cdot 3$ is better than $P + n \cdot A$,

and the conclusion the proof delivers is

Very Repugnant Addition* (VRA*). For any level $z < 0$ and any $m \geq 1$, there is a population P such that for any $n \geq 1$ there is N with $P + m \cdot z + N \cdot 3$ better than $P + n \cdot A$.

The difference is that in VRA* the background P may depend on z and m . (In Thomas's fixed-threshold vocabulary, and for a complete ordering, VRA* is equivalent to the denial of the condition WQA* below; for an incomplete ordering the denial of WQA* is weaker, since WQA* asks for positive comparisons.) Thomas writes that VRA* “is slightly weaker than VRA, because P is allowed to depend on z and m . I think that VRA* is not much less ‘repugnant’ than VRA, but then again it is not easy to understand the difference between them” (Thomas 2016, 11). Thornley's strengthened condition, which entails the denial of VRA*, is:

Weak Quality Addition* (WQA*). There are a level $x < 0$ and a number $m > 0$ such that for any population X there are a positive level u , a range $R(1, y)$ with $u > y$, and

$n > 0$ such that for any $z \geq u$ and any $B \subset R(1, y)$: $n \cdot z + X$ is at least as good as $B + m \cdot x + X$.

Thomas's corrected theorem, in his reconstruction, is that ED, GNEP, NE and WNS entail VRA^* , from which it follows that ED, GNEP, NE, WNS and WQA^* are jointly unsatisfiable. I have verified his proof, and also proved the unsatisfiability directly for the conditions in Arrhenius's exact formulations rather than Thomas's fixed-threshold ones (§6), so the repaired theorem is true in both forms. The capacity view violates WQA^* : given x and m , take X to be $m|x|$ lives at level A , so that the m lives at x are exactly covered and $I(B + m \cdot x + X) = 0$; then both sides have $J = 0$, total welfare decides, and enough lives at level 1 in B make the right-hand side better. (The baseline only helps: with $c > 0$ the misery is covered with room to spare.)

The question Thomas raises, whether VRA^* is much less repugnant than VRA , is the question whether WQA^* is as compelling an adequacy condition as WQA , and the capacity view lets us say exactly what is at stake. The two conditions agree on every axiology for which the following holds:

Compensation. For any population Y , any level $z < 0$ and any m , there is K such that $Y + m \cdot z + K \cdot 3$ is at least as good as Y .

Compensation says that any finite quantity of misery added to any population can be made up for by adding enough lives barely worth living. Total utilitarianism satisfies it, as do critical-level views with a critical level below 3 (Blackorby et al. 2005; Broome 2004), prioritarian views, and in general any view that is Archimedean in the trade between misery and marginal lives. For such views WQA^* and WQA collapse into each other, and both collapse into Quality Addition:

Proposition 2. If an axiology satisfies Compensation and WQA , it satisfies Quality Addition in Arrhenius's exact form: for any X there are a positive level u , a range $R(1, y)$ with $u > y$, and $n > 0$ such that $n \cdot z + X$ is at least as good as $B + X$ for every $z \geq u$ and every $B \subset R(1, y)$.

Proof. Given X , let x, u, y, n, m be the witnesses WQA supplies. Take any $z \geq u$ and any $B \subset R(1, y)$. Compensation applied to the population $X + B$ gives K with $X + B + m \cdot x + K \cdot 3$ at least as good as $X + B$. Since $y \geq 3$, the population $B + K \cdot 3$ lies in $R(1, y)$, so WQA gives $n \cdot z + X$ at least as good as $B + K \cdot 3 + m \cdot x + X$. Transitivity gives $n \cdot z + X$ at least as good as $B + X$. $\square$

The same argument with WQA^* in place of WQA goes through unchanged. So for any axiology satisfying Compensation, the weakening of Quality Addition to Weak Quality Addition that distinguishes the sixth theorem from the fourth buys nothing: a Compensation-satisfying axiology that accepts Weak Quality Addition, in either form, accepts Quality Addition. (The converse needs only that adding very bad lives never improves a population, which every view in the vicinity grants.) Put together with Thomas's fourth theorem, which says that ED, GNEP, NE and WNS entail Repugnant Addition, the denial of Quality Addition, this gives a corollary worth stating on its own, in Thomas's fixed-threshold setting where the two theorems share their vocabulary:

Corollary 3. ED, GNEP, NE, WNS, WQA and Compensation are jointly unsatisfiable. That is: given the four conditions that the sixth theorem shares with the fourth, Weak Quality Addition entails that

some finite quantity of misery cannot be made up for by any number of lives barely worth living.

The weakening of Quality Addition to Weak Quality Addition therefore has content only for axiologies that violate Compensation, and, given the other four conditions, it *commits* one to violating Compensation. Those are the axiologies at which the sixth theorem is aimed. They include lexical views on which the badness of a very bad life is of a different order from the goodness of a marginal one, such as the Lexical Totalism that Thomas (2018) and Thornley (2021) discuss and that Thornley (2022) elsewhere argues faces a dilemma of its own, and they include the capacity view.

Among Compensation-violating axiologies, WQA and WQA* come apart, and the capacity view is the shape of the gap. WQA* says that the quantity of misery that marginal lives cannot make up for can be fixed once, independently of the population to which the misery is added. WQA says only that for each population there is such a quantity. The capacity view takes the second option: the quantity depends on how much excellence the population already contains, because the view treats misery as redeemable by excellent lives, one for one, and as irredeemable by marginal lives. In a population rich in excellent lives, any fixed quantity of misery is covered, and once covered it is subject to the ordinary totalist trade-off against marginal lives; in a population poor in excellent lives, the same quantity is uncovered, and no quantity of marginal lives touches it.

Is that structure one an adequacy condition should rule out? I do not think the question has an obvious answer, and I want to resist two quick ones.

The first quick answer is that the capacity view is obviously absurd, so that WQA* is as harmless as Thornley says. The view has features nobody would defend as stated: the exchange rate between an excellent life and a unit of misery is arbitrary, the lexical threshold is sharp, and the view was constructed to satisfy five conditions rather than to be true. With $c = 0$ it also says that a billion lives barely worth living together with one slightly bad life is worse than no lives at all. That last consequence is worth separating from the others, because it is not a commitment of the capacity structure. The baseline parameter removes it: with $c \geq 1$ the empty population absorbs a unit of misery and the billion marginal lives then count in its favour, while every one of the five conditions continues to hold and the Very Repugnant Conclusion continues to fail for quantities of misery exceeding the baseline. Nor is the consequence peculiar to the $c = 0$ view among the sixth theorem's escapes. By Corollary 3, anyone who accepts the five conditions denies Compensation: there is some quantity of misery that no number of lives barely worth living can make up for. The $c = 0$ view denies it at every background, including the empty one. So does Lexical Totalism in the form Thornley gives it, on which "adding a combination of very negative welfare lives and barely positive welfare lives means adding (r, s) with $r < 0$ " to a value vector whose first coordinate dominates lexically (Thornley 2021, sec. 4): a very bad life subtracts a higher good that no number of barely-worth-living lives, which add only lower goods, can restore, so there too misery plus marginal lives is worse than nothing. The five conditions do not force this, as the $c \geq 1$ view shows; but the one escape from the sixth theorem in the published literature has it, and the capacity view with a positive baseline is, as far as I know, the first axiology satisfying the conditions that does not.

A second feature invites the same treatment. In the view as defined, redemptive power belongs only to lives at or above the very high level A ; lives between the very low range and A have none, so that for large A a billion lives at level $A - 1$ cannot cover a single unit of misery beyond the baseline, while for $A = 4$ “excellence” begins one step above lives barely worth living. Neither horn is forced either. The step at A is an artefact of taking the simplest potential. A graded version, in which the weight of a life rises linearly from 0 at level 3 to 1 at level A and the Non-Elitism witness is taken to be $A - 3$, satisfies all five conditions as well; I have checked this numerically for A up to 10, though the machine-checked proof is for the step version. On the graded view every life above the barely-worth-living range contributes to redeeming misery, in proportion to how far above it lies, and only the barely-worth-living lives contribute nothing. That is the version of the view that the gloss “excellence redeems, numbers do not” properly describes, and in that form the structural idea is not absurd. It is close to the intuition that the Very Repugnant Conclusion is worse than the Repugnant Conclusion: someone who accepts that enough marginal lives can outweigh a population of excellent lives, but denies that they can outweigh it when a quantity of misery has been thrown in, is treating misery as a dimension of value that marginal lives do not reach. The capacity view is the simplest way of making that attitude consistent with the four other conditions, and the fact that it must let the uncompensable quantity of misery depend on the background is a discovery about the attitude, not an artefact of the construction.

The second quick answer is that VRA^* is as repugnant as VRA , so the weakening does not matter. Here the capacity view gives a reason to disagree with Thomas’s first instinct, though I should be clear about whose reason it is. VRA^* says that for any given quantity of misery there is a background against which adding that misery together with enough marginal lives beats adding excellent lives. On the capacity view the background that does the work is one rich enough in excellent lives to cover the misery, and what VRA^* then asserts is that, in a population whose excellent lives already redeem its misery, enough additional marginal lives beat additional excellent lives. That is the Repugnant Conclusion again, stated for a population with some covered misery in it, and someone who has accepted the Repugnant Conclusion has no new reason to find it repugnant. That is how the capacity view describes its own verdict. A critic of the Very Repugnant Conclusion who is not a capacity theorist will describe the same verdict differently: in some sufficiently good world, adding lives of torment together with enough lives barely worth living is better than adding excellent lives, and the goodness of the surrounding world does not make the lives of torment less tormented. I do not think the capacity view has an answer to that critic that does not presuppose the view. What it has is a diagnosis of what the critic is committed to. The critic’s complaint is against VRA^* , which the four conditions entail; so the critic must reject one of ED, GNEP, NE and WNS, and Weak Quality Addition in any form will not help, because the critic’s complaint is about the background-relative conclusion that the four conditions reach without it. The sixth theorem, repaired, tells the critic of the Very Repugnant Conclusion that the fight is elsewhere. The unrepaired theorem told the critic, wrongly, that there was no consistent position to fight from.

So the corrected sixth theorem is a theorem about background-relative repugnance. What ED, GNEP,

NE and WNS jointly force is that the Repugnant Conclusion holds relative to any background that has redeemed its misery. They do not force the Very Repugnant Conclusion, and they do not force it relative to a fixed background. Anyone who wants the stronger result must add WQA* as a premise, and the only thing WQA* adds to WQA is the stipulation that the quantity of misery which marginal lives cannot make up for be the same for every background. Proposition 2 shows that this stipulation has no work to do against Archimedean views, which accept or reject both conditions together with Quality Addition; its content is the exclusion of views on which the uncompensable quantity of misery depends on the background, of which the capacity view is the simplest instance.

Two remarks on scope. First, the result does not depend on the denial of Finite Fine-Grainedness. The known counterexamples to the sixth theorem, Lexical Totalism among them, work by giving welfare levels a lexical structure, so that no finite sequence of slight steps leads from a very high level to a very low one; Thornley says plainly that “the theorem is only true given Finite Fine-Grainedness” (Thornley 2021, sec. 3), and he is right about the strengthened theorem. The capacity view lives on Arrhenius’s integer levels, where every level is finitely many slight steps from every other, and it satisfies the original conditions there. The lexical structure in the view is in the ranking of populations, not in the ranking of lives, and the four conditions other than WQA, which are the ones that use fine-grainedness, do not reach it. Second, the capacity view also satisfies several conditions not in the theorem: Dominance Addition, Non-Sadism, and Mere Addition among them. I note this only to forestall the thought that the view satisfies the five conditions by being pathological elsewhere. Where it is pathological is where it was designed to be, in the lexical priority of uncovered misery.

5 How the counterexample was found

The capacity view was not found by trying candidate axiologies against the conditions. It was found by a search that I think is worth describing, because it applies to any impossibility theorem of this type, because it explains why the sixth theorem fails where the others do not, and because it is cheap enough that it should be run on any new theorem of the kind.

Arrhenius’s conditions, other than the Quality conditions, are *transfer conditions*. Each says that a population obtained from another by a specified change, moving some lives between levels or adding some lives, is at least as good as the original, possibly strictly, possibly only when the background satisfies a restriction. Read as licences, they say which transfers an axiology must count as non-worsening. An impossibility proof is a chain of licensed transfers from one population to another, with a licensed strict transfer somewhere in the chain, ending at a population that the conclusion’s denial says is no better than the start. The conclusion’s denial is the odd condition out: it is not a transfer licence but a demand that certain comparisons go a certain way, and an axiology satisfies it by *refusing* a comparison that the chain would otherwise force.

This suggests looking for a quantity Φ on populations that no licensed transfer lowers, and building the axiology that ranks populations first by $\min(0, \Phi)$ and then by total welfare. If Φ is additive over union, with a weight w_ℓ for each welfare level ℓ , then backgrounds drop out and each transfer

condition becomes a linear inequality on the weights: General Non-Extreme Priority with witness G requires $w_{z-1} + Gw_x \geq w_z + Gw_y$, Non-Elitism with witness n requires $(n+1)w_{x-1} \geq w_x + nw_y$, Weak Non-Sadism with witness D requires $mw_x \geq Dw_z$, and Egalitarian Dominance requires the weights to be non-decreasing in the level. Two further constraints come from the lexical layer. One transfer lowers every candidate Φ of this kind: Non-Elitism at the very high level, when the lives being raised have non-negative welfare, removes an excellent life and adds nothing in the negative range. That transfer is licensed only in backgrounds restricted to levels at or above the raised lives', hence to populations with no misery, so it can be neutralized by capping the lexical layer at zero, provided every population without negative lives has $\Phi \geq 0$, which means $w_\ell \geq 0$ for $\ell \geq 0$. That is why the layer is $\min(0, \Phi)$ rather than Φ . And refusing a conclusion of the form “for all $N, m \cdot z + N \cdot 3 + P$ is better than $n \cdot A + P$ ” requires $\Phi(m \cdot z + N \cdot 3) < 0$ for all N , hence $w_3 \leq 0$ and so, with monotonicity and the cap, $w_0 = w_1 = w_2 = w_3 = 0$, together with $w_z < 0$.

For fixed witness sizes this is a linear feasibility problem in a dozen variables, and a general-purpose solver settles it instantly; the script that sets it up and runs it, for all six theorems, accompanies the paper. For the conditions of the sixth theorem, with every witness set to 1, it returns $w_\ell = \ell$ for $\ell < 0$, $w_\ell = 0$ for $0 \leq \ell < A$, and $w_\ell = 1$ for $\ell \geq A$: that is, $\Phi = H + T^-$, the capacity view. For the conditions of the fifth theorem, which replace Non-Elitism by General Non-Elitism and Weak Non-Sadism by Dominance Addition, the system is infeasible for every small witness size, because the unrestricted background lets the transfer at the very high level occur in populations with uncovered misery, where it must not lower Φ , and that forces $w_A \leq 0$, which the remaining conditions then propagate into a contradiction with the refusal. For the first four theorems the method is uninformative, and the reason is worth stating correctly. Their conclusions involve no misery: they say that enough lives at level 3 beat any number at level A , against a fixed background. Since total welfare grows without bound as lives at level 3 are added, an axiology of the shape under consideration can refuse such a conclusion only through the lexical layer, which requires $\Phi(N \cdot 3)$ to go to $-\infty$, that is, $w_3 < 0$; and the cap requires $w_3 \geq 0$. So no axiology of this shape refuses a misery-free conclusion, whatever the transfer conditions, and the system is infeasible before the conditions are consulted. The sixth theorem is the only one of the six whose conclusion involves misery, and the fifth is the only other one that could in principle be refused this way; there the transfer conditions do the excluding. The search finds the gap in the sixth theorem because the gap is there, and finds nothing in the fifth because General Non-Elitism closes it (the system with General Non-Elitism removed and Non-Elitism restored is feasible again).

The method has limits. It finds countermodels of one shape, lexical in an additive potential and totalist below it, and an impossibility theorem can fail for other reasons; infeasibility of the system for small witnesses is evidence rather than proof that no countermodel of this shape exists, since the witnesses are existentially quantified and larger ones might help. What the method does deliver, when it succeeds, is a candidate whose verification is routine; when it fails on a conclusion of the right shape, as for the fifth theorem, it shows which condition does the excluding. In this case it delivered the capacity view in the first run, and the verification took longer than the search.

6 Machine verification

Impossibility theorems in population ethics are proved by long chains of inequalities with many quantifier dependencies, and the error in the sixth theorem is a quantifier-dependence error: a witness chosen for one background is used for another. This is the kind of error a proof assistant catches, and the kind that referees do not. Social choice theory learned this some time ago: Arrow’s theorem and its relatives have been formally verified (Nipkow 2009), and SAT solvers have been used both to re-prove known impossibility theorems (Tang and Lin 2009) and to discover new ones (Geist and Endriss 2011; Brandt and Geist 2016). Population axiology has, so far as I can find, one mechanized result, a formalization of Parfit’s Mere Addition Paradox in a deontic logic (Parent and Benz Müller 2024), and none of Arrhenius’s theorems had been checked. I have therefore formalized the results of this paper in Lean 4 (Moura and Ullrich 2021), using only the core library, so that the development can be checked by anyone with the Lean toolchain and no further dependencies. The source accompanies the paper. Three things are verified.

First, the capacity view. Populations are represented as finite lists of integer welfare levels, and the five conditions are stated exactly as in §2, with “ $A \subset W_x, N(A) = n$ ” rendered as a list of n copies of x and “ $B \subset R(1, y)$ ” as the hypothesis that every member of B lies between 1 and y . The theorem `sixth_theorem_countermodel` states that for every $A \geq 4$ and every baseline $c \geq 0$ the view is complete and transitive and satisfies `EgalitarianDominance`, `GeneralNonExtremePriority`, `NonElitism`, `WeakNonSadism` and `WeakQualityAddition`, and that it does not satisfy `WeakQualityAdditionStar`. The proof is the case analysis of §3, mechanized; it is about 250 lines.

Second, Thomas’s reconstruction and the repaired theorem. Here populations are represented as count functions from levels to numbers of lives, with finite support, which makes the reordering of unions in the chain proofs trivial; conditions are stated in Thomas’s forms, with “very high” fixed at A and “very low positive” at $\{1, 2, 3\}$, for an arbitrary reflexive transitive relation on distributions. The theorems `T1` through `T5` and `T6star` verify that Thomas’s six theorems hold as he states them, the last being that Egalitarian Dominance, General Non-Extreme Priority, Non-Elitism and Weak Non-Sadism entail VRA^* ; `prop2` and `cor3` verify Proposition 2 and Corollary 3 in the same vocabulary. The two lemmas on which the last three rest, that General Non-Extreme Priority entails Sufficient Tradeoffs and that (General) Non-Elitism entails (General) Inequality-Averse Addition, are proved by the double inductions that Thomas’s recursive definitions of the witnesses require. Thomas’s statements and proofs are correct in every case; the formalization needed no repairs beyond making explicit the finiteness and range side-conditions that his prose leaves implicit.

Thomas’s conditions differ from Arrhenius’s exact ones in both directions: his General Non-Extreme Priority fixes the high and low thresholds where Arrhenius lets them be chosen for each level, and takes the low lives to be at a single level where Arrhenius allows a mixture. So the theorem `T6star_exact` repeats the argument for the exact conditions, with the thresholds chosen inside the conditions and the low population given as an arbitrary list of levels in the range, and derives the

negation of Weak Quality Addition* in its exact form. Its proof differs from Thomas's in two places: the lives raised to very high welfare in different applications of General Non-Extreme Priority must be sent to a common level, chosen above every threshold the condition returns, and the excellent lives in the background and in the comparison population may sit at two different levels, which are merged by an extra application of Inequality-Averse Addition before the main one.

Third, fidelity. The formal statements follow Arrhenius's text, with three deliberate choices recorded in the source. Arrhenius requires a welfare range to have at least three levels, so the range $R(1, y)$ carries the side condition $y \geq 3$, which his prose statements leave implicit. In Egalitarian Dominance the common population size is required to be positive, since with empty populations the condition would say that the empty population is better than itself. And in Weak Non-Sadism, where Arrhenius writes only "a number of lives," the countermodel is shown to satisfy the condition with a positive number of very negative lives, which is the stronger claim, while the theorems assume only the weaker form in which that number may be zero. Since the countermodel is verified against the exact conditions and the corrected theorem is verified against both Thomas's and the exact conditions, there is no tension between a verified counterexample and a verified theorem: the counterexample is to the original conclusion, and the theorem has the corrected one.

All proofs are complete, contain no `sorry`, and depend only on the three standard axioms of Lean's kernel (propositional extensionality, quotient soundness, and choice). The whole development is about twelve hundred lines. The graded variant of §4 and the solver of §5 are not part of it.

7 Conclusion

Arrhenius's sixth impossibility theorem is false. The capacity view, which ranks populations first by whether their excellent lives cover their misery and then by total welfare, satisfies all five of its conditions in their exact formulations, on integer-indexed welfare levels, while accepting the Repugnant Conclusion and rejecting the Very Repugnant Conclusion. The theorem becomes true when the Weak Quality Addition Condition is strengthened so that the uncompensable quantity of misery is fixed before the background population, as Thomas and Thornley have proposed, and the strengthened theorem has been machine-checked along with Thomas's reconstructions of the other five. But the strengthening is not free. For every axiology on which misery can be compensated by lives barely worth living, Weak Quality Addition in either form is equivalent to the Quality Addition Condition that the sixth theorem was meant to weaken; and for the axiologies on which it cannot, the strengthened condition rules out, by stipulation, the view that how much misery a population can absorb depends on how much excellence it contains. What the four remaining conditions actually establish is that the Repugnant Conclusion holds relative to any background that has redeemed its misery. Whether that is repugnant is a question for people who have made up their minds about the Repugnant Conclusion itself; it is not a new result about misery.

The broader lesson concerns how results of this kind should be checked. The error in the sixth theorem survived publication in two venues and five years of citation before Thomas found it, and it has

survived a further decade since, because finding a gap in a proof and finding out whether the claim is true are different tasks, and nobody undertook the second. It was a quantifier dependence that a proof assistant rejects in seconds, and the countermodel is one that a systematic search for lexical potentials locates in minutes. Population axiology has accumulated a stock of impossibility theorems whose premises are each the product of careful philosophical judgment and whose proofs are each a page of inequalities that almost nobody re-derives. The judgment cannot be automated. The inequalities can, and should be.

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